Activity 1: Static Hazard Hunt at NovaSense Incoming Inspection

Mapping: K1 · A1
Objective: Identify credible charge sources, exposed ESDS items, isolated conductors and field-coupling paths, then prioritise controls.

Scenario

NovaSense receives MEMS sensor trays beside a label printer, stretch-wrap station and wheeled inspection trolley. Failure returns increased after a layout change, but no one has mapped charge-generation and induction risks.

Roles

- ESD engineer
- incoming-inspection supervisor
- quality observer

Detailed procedure

1. Read the scenario and mark every ESDS item and handling handoff on the supplied floor sketch.
2. List each contact-and-separation event, material pair and motion that can generate charge.
3. Circle conductors that may be isolated from ground, including the trolley and metal fixtures.
4. Mark essential and non-essential insulators within the handling zone.
5. Trace one direct-contact discharge path and one field-induction path to a sensor tray.
6. Rate each hazard for sensitivity, exposure, detectability and consequence using the worksheet scale.
7. Select the top three risks and justify the ranking with observable evidence.
8. Propose one elimination, one grounding/bonding and one administrative control.
9. Define the measurement needed to confirm each proposed control.
10. Conduct a peer challenge: another group must identify one missed pathway or weak assumption.
11. Revise the map and record the change made after peer review.
12. Submit the completed hazard map, risk register and three-control proposal.

Questions

- Which hazard could damage a device without a visible spark?

- Which metal object could remain charged despite being conductive?
- What evidence would distinguish induction risk from direct contact?

Evidence to submit

- Annotated hazard map
- Risk-prioritisation worksheet
- Three-control proposal
- Peer-review note

Acceptance criteria

- [ ] At least eight credible hazards are identified
- [ ] Both direct and field-induced pathways are shown
- [ ] Top risks have evidence-based ratings
- [ ] Controls follow the ESD control hierarchy

Debrief

Explain which evidence changed your first assumption, which control is most reliable, and what follow-up measurement is needed.